ARCHIT AGRAWAL

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PROFESSIONAL EXPERIENCE

EdPlus, Arizona State University **06/2026 - Present**
AI Software Engineer *Phoenix, AZ*

- Designed a serverless data lake on S3: a narrow fact table plus Lance payloads queried in-process by DuckDB, so new datasets union without a migration.
- Built schema-inferring ingestion for any table shape, from crosstabs to weighted surveys and UTF-16 files, emitting each decision as a JSON recipe that replays exactly.
- Traced silent row loss in Step Functions runs that reported success and fixed it with partitioned writes and fan-in across 64 parallel lanes.
- Tracked data lineage with four typed relations (relabel, recode, wave, wave_recoded), so a second labeling pass can never pose as a trend.

MyStage Music Inc. **03/2025 - 06/2026**
Founding AI/ML Engineer *Remote*

- Scaled ingestion to 70,000+ records daily from 650+ venues using Playwright, transactional locking and entity resolution.
- Made ingestion retries safe with deterministic task IDs and atomic create-if-absent writes, preventing duplicate jobs from overwriting in-flight work.
- Built a Cloud Scheduler reprocessing loop that refreshes the most-stale sources first, tracking reprocess time and count per entity.

EdPlus, Arizona State University **09/2023 - 05/2025**
Software Engineer *Phoenix, AZ*

- Reduced transcript-analysis time from four hours to 15 minutes using a Neo4j knowledge graph and validated Cypher queries.

Knowledge Exchange for Resilience, Arizona State University **06/2024 - 08/2024**
Data Research Aide *Phoenix, AZ*

- Built an idempotent ETL pipeline for 10,000+ researcher profiles daily, using PostgreSQL natural-key upserts to prevent duplicates after partial failures.
- Put data-quality gates in PostgreSQL, with uniqueness, not-null and range constraints, so bad rows fail at ingestion; dataset accuracy rose to 95%.

Zeus Learning **01/2022 - 07/2023**
Software Engineer *Mumbai, India*

- Migrated a C#/.NET monolith to Dockerized Kubernetes microservices, reducing infrastructure cost 20% and resource footprint 35%.

PROJECTS

Image Recognition as a Service **01/2024 - 02/2024**

- Built an AWS EC2/SQS image-processing service with queue-based scaling, asynchronous results and idempotent workers for safe message retries.

TECHNICAL SKILLS

Languages & Storage: Python, SQL, TypeScript, S3, PostgreSQL, DuckDB, Lance, Firestore, Redis

Pipelines & Orchestration: ETL, AWS Step Functions (Distributed Map), Lambda, SQS, Cloud Run Jobs, Cloud Scheduler, Playwright

Data Quality & Ops: Data validation, data lineage, schema inference, idempotent reprocessing, entity resolution, CI checks, Docker, pytest

EDUCATION

Arizona State University (ASU) **08/2023 - 05/2025**
M.S., Computer Science | GPA: 4.0 *Tempe, AZ*

Dhirubhai Ambani Institute of Information and Communication Technology **08/2018 - 05/2022**
B.Tech., Information and Communication Technology *Gandhinagar, India*